Academic Background

I hold a BSc in Medical Physics with Physics from UCL, where I gained a solid foundation in theoretical physics, mathematical tools, and programming languages, alongside practical experience in medical research. My studies involved a comprehensive exploration of imaging techniques such as ultrasound, MRI, X-rays, and optics, as well as physiological recording methods that span various levels of intrusiveness and temporospatial resolution. For my final project, I developed a wearable device aimed at improving muscle control through electrophysiological signal decoding, incorporating auditory biofeedback. This project not only enhanced my engineering expertise but also deepened my understanding of how technology can be applied to improve human health, sparking my interest specifically in exploring neural function and technological applications for treating dysfunction.

Building on this foundation, I pursued a master's degree in Biomedical Engineering with a specialisation in Neurotechnology at Imperial College London. This programme significantly expanded my knowledge of fundamental neural mechanisms, including networks for cognitive functions and motor performance. Through a detailed exploration of sensory systems, including signal processing in the visual and auditory systems, I examined their function and how dysfunction arises at various biological levels and processing stages. Additionally, my practical work involved neural network modeling, ranging from single neuron properties and signal integration to learning and synaptic plasticity, as well as biomedical data analysis techniques. This experience enhanced my programming skills and provided valuable insights into experimental data interpretation. During my master's study, I worked on research projects that focused on neural signal decoding to enable effective brain-machine interface applications and tracking neural activities of in vitro models through the early developmental stages using two-photon microscopy. These real-life projects provided me with a deeper understanding of neural function and its applications in both theoretical and practical contexts.

Research Experience

Most of my laboratory-based research is focused on neurotechnology, utilising a variety of techniques including electrophysical recording, immunohistochemistry, advanced microscopy, and computational modeling. I have experience working with data collected from both human participants, animal models, and in vitro models. To advance engineering designs applied in these projects, I collaborated closely with experts in electronic and mechanical engineering, gaining hands-on experience in electronic circuit design, microcontroller programming, and 3D modelling.

Some of my key projects include:

Biofeedback with Electromyography Sonification and Music: I designed a wearable device to record EMG signals from the forearm, processing and translating these signals into audio signals for auditory biofeedback to aid in muscle control improvement. The prototype included a sensor with attached electrodes, an external signal filtering circuit, and was integrated with a microcontroller for programming. The real-time EMG signals were recorded and processed using Arduino, where the muscle signal features were decoded and mapped for auditory feedback generation.

Characterisation of a Multi-User Ultrasound Therapy System: I worked on validating a newly launched GUI for an ultrasound-guided high-intensity focused ultrasound (HIFU) system. I designed and conducted calibration experiments using various ultrasound imaging phantoms and pressure sensors, and wrote a new user guide for the system.

Tracking Changes in Brain Organoid Spontaneous Activity Over Maturation: This project aimed to investigate the impact of temperature on the neural dynamics of iPSC-derived cerebral organoids and to engineer solutions for better mimicking the brain's internal regulatory environment in vitro. I developed a temperature regulation system embedded in an imaging chamber to track neural activities in organoids. The objectives were to analyse calcium dynamics across various developmental stages, providing insights into maturation features and early neural computation, and to validate the potential of iPSC-derived brain organoids as a model for studying neurodegenerative diseases.

Additionally, I have worked on programming projects with various biomedical data, including spike train decoding for arm movement trajectory prediction, simulating models to represent nervous system mechanisms, and neuroimaging processing.

Interests and Vision

I have a strong interest in cognitive neuroscience, particularly in understanding the motor system and its functions, which align closely with the research themes in your lab. My enthusiasm lies in uncovering fundamental neural mechanisms across different levels, and I am especially drawn to studying motor function. As motor output provides valuable insights into brain function, by reflecting the processing of sensory inputs and their integration within higher-level brain regions. The complexity of precise motor control, with its time-sensitive demands and adaptability, makes it a fascinating avenue for exploring brain plasticity.

After reviewing the available project ideas for next year, I am particularly excited about the project on '**Brain plasticity for body augmentation'**. With my equipped knowledge in multi-modal neural data recording techniques, and experience in applying computational methods to decode neural signals for brain-machine interfaces along with engineering application, I see great potential to contribute to ongoing research focused on hand function augmentation, with the innovative device, **Third Thumb**. I am passionate about advancing therapies and treatment solutions for neurological diseases and disabilities, ultimately improving the patients' quality of life.

The vast potential of human brain plasticity fuels my passion for this field, and the multidisciplinary nature of your research environment excites me. My background in physics, biomedical engineering, and neuroscience positions me well to contribute to collaborative, cross-disciplinary projects, and I am eager to further refine my skills to shape my future research endeavors.